LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (AAT) (31)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Making of different jobs using Sheet metal i.e Butt/Lap joints
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Making of different jobs using Sheet metal i.e Butt/Lap joints
5. Specific Objectives	31.1.3.1 Trainee will be able to perform various repair works on aircraft structure

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration and supervised practice of Butt and Lap joint fabrication on sheet metal blanks; Safety precautions include PPE usage, tool handling, and fire safety; Instructor supervision ensures correct alignment and fastening techniques.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the two primary types of sheet metal joints used in aircraft structures?
(b) Why is proper joint selection critical for aircraft structural integrity?
(c) List three safety precautions required before starting sheet metal fabrication.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition and application of Butt joints • Edge preparation for Butt joints • Fastening methods for Butt joints • Inspection criteria for Butt joints	L & Ds	PPT, IB	15 min
(b)	• Definition and application of Lap joints • Overlap dimensions and edge finishing • Riveting patterns for Lap joints • Common defects in Lap joints	L & Ds	PPT, PDF	20 min
(c)	• Quality control checks for sheet metal joints • Repair techniques for damaged joints • Documentation of joint fabrication • Final inspection and acceptance criteria	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between edge preparation for Butt joints versus Lap joints.
(b) Describe the inspection criteria used to verify the quality of a completed Lap joint.

12. Home Assignment and Follow Up	Prepare a brief report outlining the steps required to repair a damaged Butt joint on an aircraft structure, referencing the procedures covered in this lesson and anticipating the removal of loose rivets discussed in LP 07.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Removal of loose rivets structural repair on aircraft structure	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min